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Smallpox Vaccination: A Review, Part II. Adverse Events

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Smallpox vaccination of health care workers, military personnel, and some first responders has begun in the United States in 2002–2003 as one aspect of biopreparedness. Full understanding of the spectrum of adverse events and of their cause, frequency, identification, prevention, and treatment is imperative. This article describes known and suspected adverse events occurring after smallpox vaccination.

In part I of this series, we reviewed routine use of smallpox vaccine in primary vaccination and revaccination and outlined the normal, expected reactions [1]. In part II, we detail specific characteristics, rates of occurrence, pathogenesis, diagnosis, and treatment for adverse events that occur in some individuals after smallpox vaccination.

Vaccination is generally safe and effective for prevention of smallpox. Smallpox vaccination can result in adverse reactions in individuals with and without specific, preexisting susceptibilities [2, 3]. Some of these reactions are benign, if frightening in appearance; some are serious but treatable; and a few are rare and serious and can be life-threatening. Table 1 lists adverse events that may occur after smallpox vaccination.

Vaccination can be avoided in some individuals by identifying susceptibilities (for a full discussion, see [4]). The factors known or thought to predispose to adverse events in the vaccinee or in close contacts of the vaccinee include pregnancy or breast-feeding; extensive skin eruptions present at the time of vaccination; atopic dermatitis (active or history of atopic dermatitis or "eczema"); presence or probability of T cell immune defects or disease, congenital or acquired; immunosuppressive therapy; inflammatory or disruptive diseases of the cornea or surrounding structures; and age <1 year. Antibody deficiency generally has not led to adverse events, except in a very few instances, usually in association with temporary loss of T cell function (this is discussed fully in Specific Adverse Events). Nevertheless, such individuals should not be vaccinated in situations in which no risk of contact with smallpox is present.

In the 1960s, cardiac complications after smallpox vaccine were not thought to be significant, although they were reported in some European literature. The emergence of myopericarditis in a number of individuals who received vaccine as part of the vaccination of civilians and military personnel in 2002 has led to the belief that these events truly are adverse outcomes of smallpox vaccination [5]. Most of these patients had very mild disease and returned to normal activities within 7–10 days. Whether long-term consequences exist is not known. None have been identified so far, and no susceptibility criteria are known.

On the other hand, some deaths resulting from coronary artery–related disease occurring within 1 month after vaccination appear to be only temporally related to vaccine, rather than being caused by the vaccine itself. The CDC has cautiously recommended that individuals at increased risk of cardiac disease be granted medical deferrals for smallpox vaccination.

The rate with which each of the adverse events occurs can only be estimated. A series of studies conducted in the 1960s yielded approximations of the frequency for some adverse events (table 2) [6–9]. These estimates have been reanalyzed

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Table 1. Adverse events that may occur after smallpox (vaccinia virus) vaccination.

Adverse event	Event subtype
Noninfectious rashes (erythema multiforme)	Macular
	Vesicular
	Pustular
	Urticarial
	Mixed
	Stevens-Johnson syndrome
Bacterial infection	Staphylococcal
	Streptococcal
	Enteric
	Anaerobic
	Mixed
Inadvertent inoculation	Autoinoculation
	Contact inoculation
	Vaccinia keratitis
	Eczema vaccinatum
Congenital vaccinia	Fetal/neonatal disease
Generalized vaccinia	Single occurrence
	Recurrent disease
Progressive vaccinia	In congenitally T cell–deficient individuals
	In antibody-deficient individuals (rare)
	In individuals with any disorder that suppresses T cell function (e.g., cancer and HIV infection/AIDS)
	Accompanying T cell–suppressive therapy (e.g., high-dose steroids and some forms of chemotherapy)
Postvaccinial encephalitis	Variable severity
Myopericarditis[a]	
From questionable case reports (temporal association only)	
Hemolytic anemia	
Arthritis	
Osteomyelitis[b]	
Cardiac lesions	
Thrombocytopenia	

[a] No known precursors at present.
[b] Vaccinia virus has been isolated from bone in some instances.

for contact vaccinia [10]. These studies were limited by the nature of reporting, the variability in the definitions used for each adverse event, and the incomplete nature of retrospective clinical data, but they provide our best estimates of postvaccinal morbidity. The current population differs from that in the period from the 1950s through the 1970s; there are more immunosuppressed individuals, more persons who have never been vaccinated, and more persons with atopic dermatitis. Screening vaccinees and their contacts may reduce the resultant potential for a greater frequency of certain adverse events but cannot eliminate it entirely. This is especially true if the United States uses widespread vaccination over a short period of time.

ACCIDENTAL ADMINISTRATION BY THE ORAL OR PARENTERAL ROUTE

Occasionally, individuals may ingest vaccine accidentally or may be mistakenly injected by the subcutaneous or intramuscular route [2, 11]. No harm has been recorded from such events, but it is prudent to observe the individuals clinically for 1–2 weeks (V.A.F., personal experience). If the injection is subcutaneous, some individuals may respond with a normal cutaneous response and may develop serum antibody, whether there is a skin response or not; however, the response is inferior to that observed with skin vaccination, and this method of vac-

Table 2. Estimated rates of occurrence of adverse events associated with smallpox (vaccinia virus) vaccination.

Type of adverse event	Rate of adverse events occurring after vaccination at indicated age, no. of events/1 million primary vaccinees <1 year	1–4 years	5–19 years	⩾20 years
Death (all causes)	5	0.5	0.5	5[a]
Postvaccinial encephalitis	6	2	3	4
Vaccinia necrosum	1	0.5	1	>10[a]
Eczema vaccinatum	14	44	35	30
Generalized rashes[b]	400	9600	140	250
Accidental implantation	507	577	371	606

NOTE. Serious reactions are rare, numbering 3–5 events/1 million primary vaccinations. The small numbers given in this table, divided by age group, may exaggerate the actual frequency. Data are from [6–9].

[a] These nos. are "modern" estimates based on the change in population.

[b] See "Erythema Multiforme."

cination is not recommended [12, 13]. Intramuscular injections of this highly dermatotrophic virus do not result in harm or immunization (V.A.F., personal experience).

SPECIFIC ADVERSE EVENTS

The images shown in this report represent selected instances of adverse events that we believe are typical. Many more images, showing a fuller range of manifestations, can be found on the CDC Web site (http://www.bt.cdc.gov/training/smallpoxvaccine/reactions).

"Erythema Multiforme"

General. Many vaccinees develop erythematous papules, plaques, or urticaria-like skin rashes after vaccination, almost all of which are benign, though occasionally frightening in appearance. In previous reports, these rashes have been, for the most part, collectively termed or reported as "erythema multiforme," although it is likely that many do not fulfill the histopathologic criteria for erythema multiforme [14]. The rashes are either toxic or hypersensitivity reactions and require only symptomatic therapy. Rarely, there can be evolution to Stevens-Johnson syndrome, with mucosal involvement; in such cases, aggressive hospital-based management is required.

The rate of occurrence of these rashes is indicated in table 2. These are very frequently seen after vaccination, particularly in children after primary vaccination, and may have multiple etiologies, including hypersensitivity to an antibiotic or other component of the vaccine. These rashes were reported to occur as frequently as 1 in 100 primary vaccinees in a prospective vaccination study in a comprehensive care clinic with a computerized record-keeping system [15]. There appear to be no predictors that can be used in screening out those who might develop such a reaction. Immunologic data on affected vaccinees is lacking, and we suggest that future instances be investigated by modern immunologic techniques.

Clinical manifestations. A range of cutaneous presentations may occur 1–2 weeks after vaccination (figure 1). The mildest consist of erythematous, irregular patches, often covering extensive body surface or limited to the arm in which the vaccine was placed. The rash may be symmetrical and often involves the palms and soles. Urticaria, vesicles, or, rarely, pustules can occur. A few individuals have extensive involvement, with more palpable lesions covering most of the body. The rash is pruritic in most instances. The degree of itching and erythema is intense. Rarely, bacterial infection in excoriated skin may occur. The vesicular form can be confused with poxvirus diseases, such as generalized vaccinia or chickenpox. Rarely, Stevens-Johnson syndrome, with full-body involvement and conjunctival and corneal inflammation, is seen. Extensive desquamation can result.

Diagnosis. The presence of vaccination and the appearance of the rash suffice to make a clinical diagnosis, and no further studies are required. The vesicular and papular eruptions may be confused with generalized vaccinia, accidental inoculation, or chickenpox or other viral exanthems. The use of the term "generalized vaccinia" to describe these rashes should be avoided, because that term applies to a very specific syndrome. Occasionally, a skin biopsy may be diagnostic. In some instances, virologic laboratory diagnosis might also be considered to aid in differentiating among these possibilities. Immunologic studies, although not required for diagnosis, may be helpful in gaining better understanding of the pathogenesis of these reactions.

Management. Most of these reactions will resolve spontaneously. For comfort, use of antihistamines or antipruritic measures is indicated. The skin requires no special care. If Stevens-


Figure 1. Examples of "erythema multiforme," a noninfectious rash commonly seen after smallpox vaccination. (Photographs from V.A.F.)

Johnson syndrome occurs, appropriate supportive measures in an in-patient setting are often necessary.

Prevention. There are no known methods of predicting or preventing these reactions. Screening for sensitivity to the specific antibiotic(s) in the vaccine or to other components included in the vaccine may help prevent some of these rashes. It is hoped that these types of rashes will occur much less frequently when the purified tissue culture vaccines under development are used.

Bacterial Infection of Vaccination Site

General. Historically, tetanus and syphilis were transmitted by variolation or by contaminated vaccinations [11, 16]. Today, staphylococci and group A β-hemolytic streptococci are the organisms most likely to cause infection in healthy individuals. Occasionally, enteric or anaerobic organisms may be encountered. In infants, fecal contamination is common. Disruption of the skin by vaccination may provide a fertile field for bacterial superinfection. In immunodeficient patients, bacterial infection is common and may be due to unusual organisms.

Mixed infections also may occur. Occlusive dressings may result in maceration of the skin, more-frequent infection, and the possibility of anaerobic infection and more-serious disease. Recent experience suggests that a doubled-over sterile gauze pad under a semipermeable covering may prevent maceration and subsequent infection.

The frequency of bacterial infections is unknown and may be in part dependent on the presence or absence of associated skin infections, such as impetigo. In one author's experience (J.M.N.), 1 patient among 500 vaccinees in a prospective series developed streptococcal infection [15]. Infants and young children experience this adverse event commonly because of unfettered scratching of the vaccination site and nonhygienic behavior. Bacterial superinfection may complicate progressive vaccinia as a consequence of both the presence of necrotic tissue and immunodeficiency.

Clinical manifestations. Staphylococcal infection is due to *Staphylococcus aureus* or related species and results in a vesiculopustular lesion at the site of vaccination, which often spreads peripherally in circumferential fashion (figure 2). Bacterial lymphangitis and regional lymphadenitis may occur, but most often the lesions are solely superficial infections. Streptococcal infection is most often caused by group A β-hemolytic streptococci and may result in lesions similar to staphylococcal impetigo, but more commonly a thick, piled-up plaque or eschar is seen at the vaccination site (figure 2). Lymphangitis and edematous painful regional lymphadenitis occur commonly. Enteric and anaerobic infections may present with foul puru-

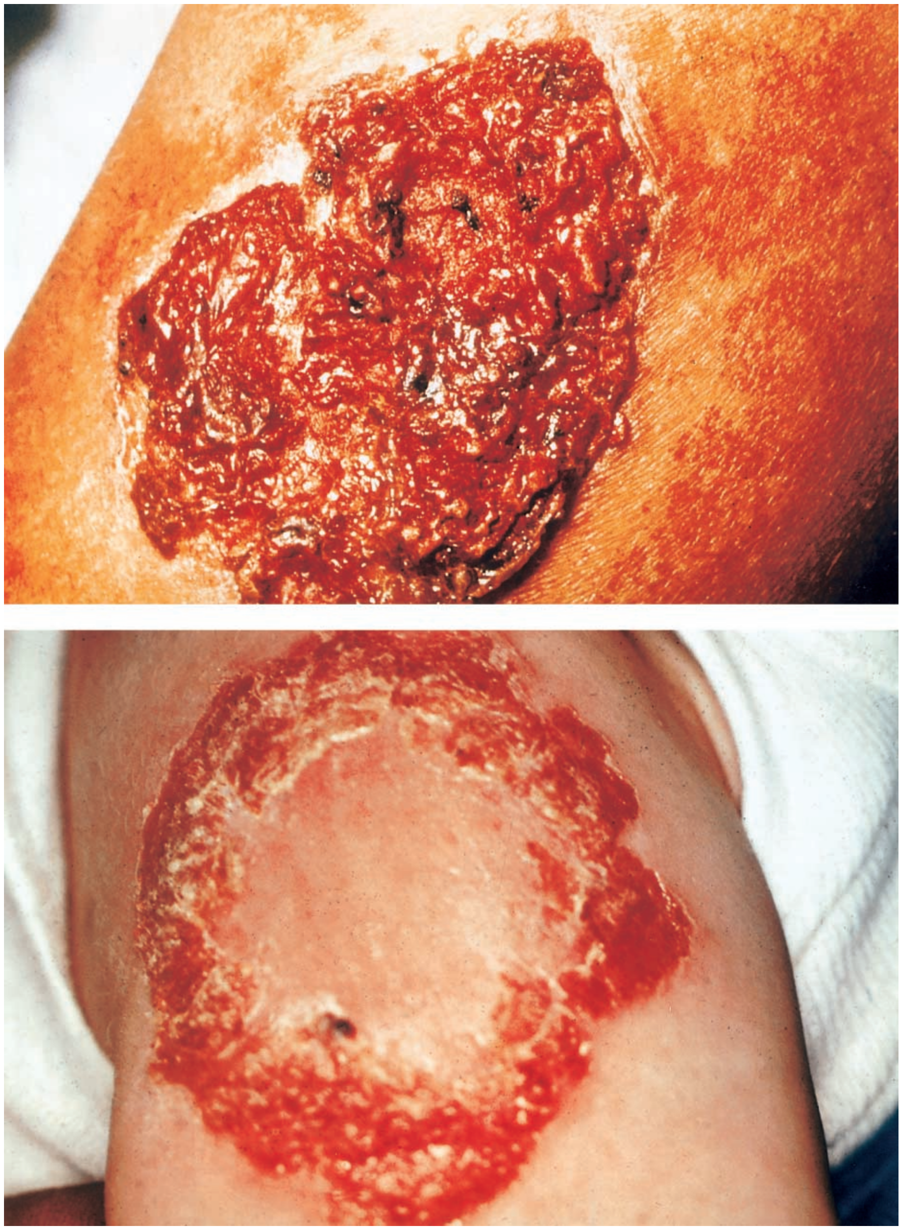

Figure 2. Bacterial superinfection of vaccination site. *Top,* Streptococcal infection (lesion is stained with mercurochrome). *Bottom,* Staphylococcal impetiginous infection. (Photographs from V.A.F.)

lence or with extensive necrosis at the vaccination site. Necrotizing fasciitis has also been encountered. One patient in a series by one of the authors (V.A.F.) developed extensive necrotizing fasciitis in contact vaccinia surrounding a recent colostomy site.

Diagnosis. Most often, bacterial infections of the vaccination site are clinically recognizable and can be confirmed by culture. Purulent lesions should be swabbed, and, if the lesions are vesicular or pustular, aspirated specimens should be obtained. The blood should be cultured if septic symptoms accompany local infection. Immunologic studies may be warranted if immunodeficiency is suspected.

Management. Appropriate antimicrobial therapy is required and should be selected on the basis of anticipated etiology and the results of culture and sensitivity testing. The area should be kept clean and debrided as appropriate. A loose dressing and topical antimicrobials may prevent spread and potentially hasten healing. In individuals with immunodeficiencies, appropriate replacement therapy should be used. Consultation with infectious diseases and immunology experts is advisable.

Prevention. Vaccination should be avoided for anyone with existing skin infection. Care of the vaccination site should be stressed to vaccinees and their families to avoid contamination. Frequent hand washing, fingernail trimming, and swathing of infants' hands to prevent scratching of the vaccination site should be used. Local or systemic use of prophylactic antimicrobials without evidence of infection is *not* recommended, because resistant organisms may then dominate subsequent infection of the site.

Inadvertent Inoculation Vaccinia (Accidental Inoculation, Implantation Inoculation, Contact Vaccinia, Autoinoculation, Eczema Vaccinatum [EV], and Vaccinia Keratitis)

General. After primary smallpox vaccination, high titers of vaccinia virus are extruded onto the surface of the site [16, 17]. Less virus is present on the skin after revaccination. This surface virus is easily transferred to the hands and to fomites. The vaccination site is frequently pruritic, and many vaccinees, particularly children, tend to scratch or otherwise make contact with the site. Virus is then transferred by touch or scratch. Minor breaks in the skin provide a fertile field for implantation. Virus is more easily transferred to abnormal skin or mucosa. Because the virus is highly dermatotrophic, a primary vaccination occurs at the site of implantation. In healthy individuals, each lesion will follow the same course as the primary vaccination. If the individual has a cell-mediated immune (CMI) defect, however, the implantation can be serious and life-threatening (see Progressive Vaccinia for detailed information).

Accidental implantation varies from single lesions to massive involvement of disruptive skin disorders, particularly atopic dermatitis [10, 11, 16, 18–21]. Infection has also occurred in laboratory workers and after accidental hand contact with a recombinant vaccinia virus [22, 23]. The degree of skin involvement parallels the severity of accidental implantation. Slight lesions, such as superficial wounds, burns, and acne, pose less of a risk than does extensively involved skin. In individuals with atopic dermatitis, even "healed" skin may be subject to inadvertent inoculation because of the underlying immune defect [24]. Adequate screening of vaccinees and contacts may prevent transmission. Covering the lesion and instructing the vaccinees about the potential for transmission may also lower the risk. Susceptible contacts of vaccinees are also at risk of contact vaccinia infections.

Young children are most susceptible to extensive inoculations because of their tendency to scratch an itching vaccination site. Older individuals implant virus frequently on the face as a result of inadvertent contamination of the hands. Lesions in eczematous skin (more correctly termed "atopic dermatitis"), in disrupted skin, and in the eye pose special hazards, because the infection can be extensive in skin lesions and can be a threat to sight when it occurs in the eye. These syndromes are discussed separately below, to emphasize the special circumstances and severity.

Clinical manifestations. Lesions due to implantation progress in the same fashion as those resulting from the primary vaccination in healthy individuals (figure 3). In patients with CMI defects, each lesion progresses without an inflammatory response, does not heal, and expands. In individuals with atopic dermatitis or disrupted skin, the lesions tend to be confluent and massive in extent. Periorbital, conjunctival, and corneal lesions are seen in persons with prior inflammatory eye disease. The periorbital and conjunctival lesions undergo the same evolution as the primary lesions in skin and mucosa, whereas the corneal infection causes ulceration and, ultimately, cloudiness.

Diagnosis. In healthy persons, the implanted lesions are characteristic of a normal vaccination and are easily identified. In most instances, laboratory tests are unnecessary. If the lesions are extensive, immunologic and virologic investigation is warranted. In individuals who lack specific contact information, it may be necessary to establish the etiology of a given lesion by viral studies.

Management. No specific treatment is required if there is only 1 or only a few implantations. If there are multiple or confluent lesions, administration of vaccinia immunoglobulin (VIG) is indicated (see the section VIG for specific details).

Prevention. Before vaccination is considered, a medical history should be carefully obtained to rule out atopic dermatitis, diseases involving disrupted skin, burns, wounds, and inflammatory eye disease in the vaccinee or his/her contacts. Vaccination of patients with small areas of skin disruption may be permissible, if accompanied by instructions about how to limit autoinoculation and by covering the primary vaccination site.

EV

General. The literature about EV confuses true atopic dermatitis with "eczema," a term loosely applied to a variety of disruptive skin disorders. Individuals with true atopic dermatitis are at great risk from implantation of vaccinia virus into the diseased skin, which sometimes has a fatal outcome [2, 6–11, 15, 18, 19, 25]. Before the advent of VIG treatment, Kempe [2] reported, in experimentally uncontrolled observations, that this disorder had a 30% mortality rate at a referral center that saw the most serious instances; after the use of VIG, the mortality rate was reduced to 7% in the same unit's experience. Copeman [19], on the other hand, reported a 12% mortality rate among 187 unselected patients, which suggests that the overall mortality in the general population was lower than that among referred patients with serious cases. Atopic dermatitis implies both a skin abnormality and a defect in immunomodulation. The term "eczema" has been applied to a variety of skin disorders; hence the term "eczema vaccinatum." Most patients have atopic dermatitis, the most serious underlying disorder; some have other generalized skin disorders, such as seborrheic dermatitis or Darier disease (keratosis

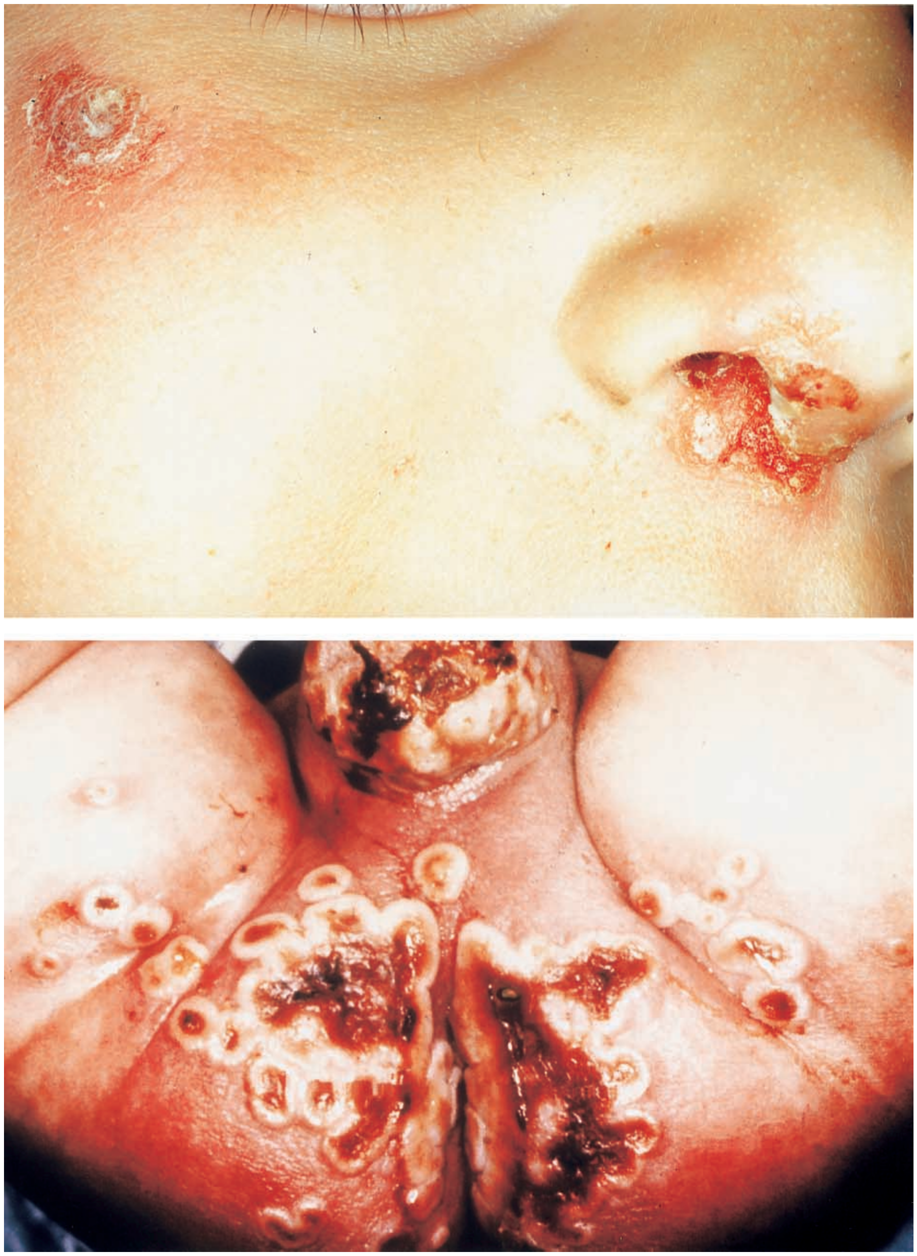

Figure 3. Examples of inadvertent inoculation with vaccinia. *Top,* Vaccinia virus transferred to nose and face in a child with allergic rhinitis. *Bottom,* Vaccinia virus transferred to diaper dermatitis from recently vaccinated parent or sibling. (Photographs from V.A.F.)

folicularis, an autosomal dominant, slowly progressive disorder of keratinization) [21, 25, 26].

Disrupted skin in patients with atopic dermatitis permits viral implantation. Once the virus is implanted (and it may be implanted at multiple sites), it spreads from cell to cell and sometimes enters a viremic phase, producing extensive lesions dependent only on the extent of the abnormal skin and other immunologic factors. An underlying T cell immunomodulatory defect is thought, on the basis of a propensity to develop cutaneous viral and fungal infections and a decreased sensitivity to contact dermatitis that is T cell mediated, to be present in some patients with atopic dermatitis [25]. In addition, some patients with T cell immunodeficiencies have atopic dermatitis as a feature, which suggests that a link exists between these 2 entities. T cell defects are a contributory factor in the severity of vaccinia infection and help explain lesions in "healed" skin.

Transfer of vaccinia virus can occur through autoinoculation or through contact with a vaccinee who has a lesion that is in the florid stages. With early recognition and appropriate use of VIG, mortality can be reduced and morbidity alleviated. The severity of EV and mortality seems to have been greater among contact cases than among vaccinees [10]. This could be explained by the greater exposure to several sites that may occur in contact vaccinia and the difficulties involved in identifying severe atopic disease in potential contacts, compared with those who present for vaccination.

Currently, there are ~27 million individuals in the United States who have atopic dermatitis (table 3). Even healed skin is not normal, and EV has occurred in such individuals at the sites of prior dermatitis. For an analysis of contact vaccinia, including EV, see Neff et al. [10]. If early diagnosis is not established and treatment with VIG is delayed, viremia can ensue, which allows the virus to spread to other parts of the body, including skin that is not affected by eczema. Bacterial and fungal invasion may occur as a late stage of untreated EV.

Clinical manifestations. The individual lesions of EV are identical in appearance to those associated with primary vaccination and undergo identical evolution. Because most individuals have large contiguous patches of affected skin, confluent lesions are usually seen (figure 4). These often cover the entire face, antecubital fossa, or area behind the knee, in the popliteal fossa. Lesions may occur as a result of autoinoculation after the initial transfer or by viremic spread. Untreated patients become quite ill, evidence systemic symptoms, and may develop septic shock and die.

Bacterial infection of the lesions may occur. The lesions are similar in appearance to those described in Bacterial Infection of Vaccination Site but are more extensive and necrotic. Bacteremia and septicemia may result from local contamination or frank infection of the site, at which time the patient may experience fever, chills, obtundation, and even coma. Abscesses may occur by extension from infected sites. Successfully treated individuals will heal, as with normal primary vaccinations. Scarring may be extensive, depending on the extent of the infected area.

Diagnosis. The history of atopic dermatitis and the typical clinical appearance of lesions, with a visible vaccination site or history of contact with a vaccinee, usually establish the diagnosis of EV. Diagnosis may be more difficult in contact cases, because history of contact with a vaccinee may be unknown or its significance unappreciated. Differentiation from herpes infection, eczema herpeticum, or other poxvirus disease may require virologic diagnosis. Immunologic studies, particularly of T cell function and IgE levels, should be performed in consultation with an immunologist, allergist, dermatologist, or infectious diseases expert to help achieve better understanding of the pathophysiology of EV. Appropriate bacterial and/or fungal cultures of skin or blood samples may be indicated if there is evidence of contamination or symptoms that suggest the presence of bacteremia or septicemia.

Management. EV demands urgent treatment with VIG. EV-associated death generally can be prevented if patients are treated as soon as reasonably possible, preferably within 1–2 days after the diagnosis is established. However, even if there is a delay in recognition, VIG therapy should be instituted promptly. In the past, the standard initial dose of intramuscular VIG was 0.6–1.0 mL/kg of body weight. As much as 5–10 mL/kg intramuscular VIG, divided into multiple doses and given over the course of several days, was administered to patients with extensive or systemic disease (see the section VIG for details).

Table 3. Estimated numbers of persons in the population currently who have conditions that might predispose to progressive vaccinia.

Predisposing condition	No. of persons
Congenital immunodeficiency (particularly of the T cell system)	Unknown
HIV infection/AIDS	900,000
Aware of HIV positivity	600,000
Unaware of HIV positivity	300,000
Receipt of organ transplant	184,000
Cancer	8,500,000
Receipt of immunosuppressive therapy	Unknown
Atopic dermatitis (current or history of)	27,000,000

NOTE. Adapted from Kemper et al. [27].

When bacterial or fungal infection is present, the choice of appropriate antibiotic treatment should be guided by knowledge of which are the most probable causative organisms (*S. aureus*, streptococci, and enteric bacteria) and by the results of culture and sensitivity testing. Septic shock requires intensive care unit (ICU)–level care and consultation with specialists, with special attention to fluid and electrolyte balance in individuals with extensive skin involvement.

Prevention. The most effective measure for prevention of EV is identification of susceptible individuals by history of or current atopic dermatitis in vaccinees or their potential contacts. Individuals with atopic dermatitis or other extensive skin disorders should not be vaccinated if there is no risk of exposure to smallpox. Healthy candidates for vaccination who have contact with persons with atopic dermatitis should not be vaccinated unless they can avoid person-to-person contact with the susceptible person(s) until the scab separates from the vaccination lesion.

If smallpox is introduced, vaccination of individuals with these predispositions or of their contacts may become necessary. The risk of contracting smallpox must be balanced against the risk of adverse events resulting from vaccination. Should an effective antiviral therapy be developed, then EV would be a logical choice for treatment. In all instances of EV, consultation with appropriate experts is strongly recommended.

Vaccinia Keratitis

General. Lesions of the cornea secondary to implantation of vaccinia virus threaten eyesight. Corneal abrasion, ulceration,

Figure 4. Characteristic rash and distribution of eczema vaccinatum in patients with atopic dermatitis. (Photographs from V.A.F.)

and subsequent clouding may result in significant impairment of vision. Vaccinia virus can be implanted into diseased or injured conjunctiva and cornea, resulting in viral replication with ulceration and, ultimately, in an antigen-antibody interaction that leads to corneal cloudiness. Ten days after transfer of the virus, the clinical signs of infection appear. There may be considerable scarring as the lesion heals, with significant impairment of vision.

Clinical manifestations. One week to 10 days after implantation of the virus in the eye, a central, grayish, discoform corneal lesion can be seen [28, 29]. Often, there are accompanying or preceding palpebral or periorbital vaccinations (figure 5) [30]. When periorbital or mucosal involvement occurs, there may be considerable pruritus, which leads to further rubbing of the eye and continued spread of the virus.

Slit-lamp examination is critical to defining the early stages of infection and to following the course of disease and response to treatment. As the infection progresses, a deeper, ringlike lesion appears in the cornea. There may be uveal involvement, Descemet's membrane may be infected, and more-distal parts of the cornea may be involved. The corneal lesions appear craterlike and are indurated, edematous, and infiltrated. Cloud-

Figure 5. Vaccinia keratitis. Infection of the cornea and conjunctiva in a mother with allergic conjunctivitis whose infant was recently vaccinated. Note the cloudiness of the cornea, which limits vision. (Photograph from V.A.F.)

iness of the cornea may occur in untreated individuals and was seen regularly in those treated with VIG, in one author's experience (V.A.F.) [31]. Cloudiness often was of sufficient size to impair vision.

Diagnosis. Consultation with an experienced ophthalmologist is strongly recommended. Slit-lamp examination should be done in all instances. Diagnosis is made on the basis of typical clinical findings. It usually is not necessary to undertake virologic studies, unless there is confusion with herpetic keratitis or other inflammatory eye disease. Lesions around the orbit are similar in appearance to typical vaccination reactions, and the diagnosis is apparent.

Management. The CDC recently convened an expert panel on ocular vaccinia; the results of that panel's deliberations can be found in [4]. Topical antiviral agents and interferon were found to be effective in treating keratitis in clinical and experimental investigations [31–36]. Which of the currently available agents is the best choice should be determined by an experienced ophthalmologist. Antiviral nucleoside analogues appear to be the best choice for viral inhibition, but the correct choice for adults and children may differ, and consultation is required. Trifluridine is one of these compounds, but it is not approved by the US Food and Drug Administration for use in treating vaccinia keratitis, although it has been shown to be effective in an animal model [37]. It can be used off-label. Vidarabine could be expected to be effective in vaccinia but is no longer marketed by the manufacturer [34].

There is some controversy about the use of VIG. In the past, this form of treatment was widely recommended on the basis of anecdotal evidence of apparent healing after administration [28, 30]. No controlled studies were performed. In the experience of one of the authors (V.A.F.), corneal clouding in humans appeared to follow the use of VIG, and such clouding limited sight [11, 31]. A controlled animal experiment was conducted; in that study, large doses of VIG contributed to excessive, widespread scarring of the cornea [31]. Limited doses of VIG and use of topical idoxuridine did not appear to be of benefit in treating vaccinia keratitis in this animal model. One experienced ophthalmologist believes that the use of 1 or 2 doses of VIG is effective and without adverse consequence (D. P. Langston, oral communication, 7 November 2002). However, the recommendation has been accepted by the Advisory Committee on Immunization Practices (ACIP) of the CDC that VIG not be used to treat vaccinia keratitis [32].

We believe, on the basis of the preponderance of evidence, that use of VIG is contraindicated for treatment of vaccinia keratitis; however, the ACIP now states that VIG is not recommended for treatment of isolated vaccinia keratitis but may be used if keratitis occurs in association with a life-threatening complication, such as progressive vaccinia. In the absence of keratitis, periorbital lesions could be treated with VIG and consideration given to use of antiviral agents instilled in the eye prophylactically. However, there are no data to support the use of antivirals in this fashion.

Other forms of treatment may be recommended by experienced ophthalmologists, but the rationale for such treatments has not been established (e.g., debridement or instillation of cortisone or antibacterial preparations into the eye). For full discussion of the ACIP position on ocular vaccinia, consult [4].

Prevention. Vaccination of individuals with inflammatory eye disease should be avoided if there are abrasions in the cornea and if steroids have been used in treating the eye disease. However, if smallpox has been introduced into the general population and the individual has been exposed, then vaccination can be done, with careful instruction to vaccinees and contacts of vaccinees to avoid touching, rubbing, or performing any maneuvers that might transfer vaccinia virus to the eye. Measures taken to contain the virus at the vaccination site (e.g., wiping excessive vaccine off the arm immediately after vaccination and the use of a semipermeable occlusive dressing) may also be helpful. Whether the use of prophylactic instillation of antiviral compounds is beneficial is unproven, and administration of such agents not recommended.

Generalized Vaccinia

General. There is confusion about the diagnosis "generalized vaccinia." Generalized vaccinia is a specific syndrome resulting from viremic spread of virus from the vaccination site in presumably healthy individuals. Despite the appearance of the lesions, it is almost always a benign complication of primary vaccination [2, 11]. Previous reports have mistaken EV, progressive vaccinia, and inadvertent inoculation syndromes for generalized vaccinia.

The frequency of true generalized vaccinia is not known, but it is believed to be rare, especially after vaccination with the New York City Board of Health vaccinia virus strain. Previous reports have overemphasized the rate of occurrence because of the lack of adherence to this strict definition. Most cases were studied at a time when immunologic knowledge was less complete than at present, and therefore the characteristics that enhance susceptibility were not known. Many of these patients may have had minor immunologic abnormalities, particularly of the immunoglobulin B cell system, or generalized vaccinia may have been confused with mild vesicular or other benign erythematous rashes that can be seen after vaccination but do not have a viremic component.

Recurrent episodes have been seen in some individuals, lending credence to the hypothesis that an immunologic defect enhances susceptibility to generalized vaccinia [2, 11]. Such a defect most likely would be an antibody or B cell deficiency or an aberrant defect, because viremia is normally controlled by the presence of antibody. The lesions of generalized vaccinia heal without incident, which suggests that T cell immunity is intact.

Clinical manifestations. Within a week after vaccination, lesions appear on unimmunized skin that appear to derive from viremia (figure 6). Lesions are similar in appearance to those associated with primary vaccination but are usually smaller and rapidly evolve to scarring, often within 5–6 days. Lesions may occur on any part of the body and are seen most often on the trunk and abdomen and less commonly on the face and limbs. Lesions may occur on the palms and soles. Rarely, the rash is profuse and covers most of the body. Even more rarely, lesions may recur at 4–6-week intervals for as long as 1 year, unless VIG treatment halts the recurrences.

Diagnosis. Exact diagnosis of generalized vaccinia can be difficult, and this disorder must be differentiated from the following:

1. Vesicular or vesiculopapular erythema multiforme. The lesions of generalized vaccinia do not umbilicate, nor do they resemble the vaccination lesions of erythema multiforme.
2. EV. A history of skin disease and the typical distribution of the lesions in EV may help differentiate these conditions. It may be difficult to distinguish between EV in healed skin and generalized vaccinia, however.
3. Early stages of progressive vaccinia. The primary lesion in progressive vaccinia is without inflammation and does not heal.
4. Severe chickenpox. The appearance and distribution of lesions in these 2 conditions differ significantly. It may be more difficult to distinguish between lesions in adult patients, particularly those with bacterial superinfection.
5. Pustular impetigo. This condition may be identified by appropriate bacterial cultures.

Differentiation between smallpox and generalized vaccinia may be very difficult, especially in partially immune patients. Vaccinia lesions will occur after vaccination, but if the patient has been exposed to smallpox, he/she may have modified smallpox. Virologic differentiation is essential in this instance. Virologic diagnosis is otherwise seldom needed. Consultation with an immunologist and appropriate studies, particularly of the B cell immune system, should be undertaken to determine whether an immunologic deficiency is present.

Management. Most instances of generalized vaccinia, particularly if the lesions are few, require no specific therapy. Patients with extensive or recurrent disease should be treated with VIG (see the section VIG for details). If an antibody (B cell)–deficient state or another immunologic abnormality is diagnosed, then both VIG therapy for the viral disease and appropriate therapy for the immunologic deficiency must be administered, in consultation with an immunologist or other specialist.

Prevention. It is not currently possible to predict which patients will develop generalized vaccinia, and no preventive measures are known. If an individual has a history suggestive of or an established diagnosis of an antibody (B cell system) immunodeficiency, that person should not be vaccinated.

Figure 6. True generalized vaccinia. Note that each viremic lesion is similar in appearance to those associated with normal primary vaccination. (Photographs from V.A.F.)

Congenital Vaccinia

General. Infection with smallpox of the fetus in utero can occur if a pregnant woman is vaccinated [38–40]. Early fetal death may result, and congenital vaccinia may supervene if infection occurs in the last trimester. This is a very rare event. Despite large-scale vaccination campaigns in the past that undoubtedly resulted in vaccination of many pregnant women, <40 cases of congenital disease have been recorded in the literature. The third trimester appears to be a critical time for risk to the fetus of congenital vaccinia, although there have been reports of vaccination earlier in pregnancy resulting in evidence of disease at birth [38]. Viremia in the vaccinated pregnant woman is thought to lead either to direct infection of the fetus or to infection of the fetus secondarily after placental or amniotic membrane infection. No proven instance of congenital abnormalities has been attributed to vaccination of pregnant women. Vaccinations of adults, including pregnant women, in the past were almost always revaccinations; today, such vaccinations would be primary vaccinations in most. Therefore, it is impossible to predict what the outcome for the fetus would be in these changed circumstances.

Clinical manifestations. The affected infant is often premature. The lesions in the newborn infant may be typical of generalized vaccinia or may be progressive in nature. Lesions are often confluent and extensive. Death occurs almost always before birth or shortly thereafter. One instance of mild disease has been recorded, in a newborn whose mother was vaccinated at ∼3–4 months gestation [40]. The authors of the report attribute the child's disease to the mother's vaccination, but no evidence is presented to verify that contention, and they point out that other causes were as likely as the vaccination to have been responsible. The infant manifested a mild skin eruption with scarring, choroiditis, macular destruction, and hypoplastic bone changes and other "deformities," which suggests that another congenital infection might have been responsible.

Diagnosis. In the absence of natural smallpox in the environment, a history of maternal vaccination and typical lesions in the infant suffice to establish the diagnosis. Virologic studies are confirmatory. If smallpox is extant, differential diagnosis is based on isolation of virus from the lesions.

Management. There is no known treatment for congenital vaccinia. VIG could be administered and antiviral therapy used, if available, but there is no experience with either of these forms of treatment [41, 42].

Prevention. Vaccination of pregnant women with smallpox should be avoided, as with any live virus, unless the woman has been exposed to smallpox, in which case she should be vaccinated.

Progressive Vaccinia (Vaccinia Necrosum, Vaccinia Gangrenosa/Gangrenosum, and Disseminated Vaccinia)

General. Progressive vaccinia is the most severe complication of smallpox vaccination [2, 3, 11, 43–47]. It is life-threatening. Various terms ("vaccinia necrosum," "vaccinia gangrenosa/gangrenosum," "generalized vaccinia," and "disseminated vaccinia") have been used to describe this syndrome. We prefer the term "progressive vaccinia." The frequency of this complication is not known with precision (see table 2 for rates). Table 3 lists the estimated numbers of persons in the present population who may be susceptible to progressive vaccinia.

Progressive vaccinia occurs because of an immune defect. Nearly all patients with progressive vaccinia have a CMI defect (T cell deficiency). Progressive vaccinia occurs in patients with CMI deficiency but intact antibody (B cell) function, but it is often limited to progression in the skin without viremic spread [3]. In such patients, antibody presumably neutralizes virus in the blood, preventing dissemination. Experienced investigators and clinicians have noted the variability in severity and manifestations of this condition, which is probably related to differences in the degree of T cell deficiency. Other subtypes of CMI deficiencies were being defined at the at the time that progressive vaccinia was seen; a T cell defect associated with an inosine phosphorylase defect was described, and patients with T cell defects and intact or partially intact B cell function also were described [2, 3, 48].

The virus multiplies by cell-to-cell spread at the primary vaccination site, causing the lesion to expand circumferentially. There is virtually no inflammation at the margins of the lesions, unless bacterial infection has supervened or immunocompetent cells have been administered (such as transfusion of fresh blood). The outer edge of the primary site is usually a ring of confluent vesicles that expands circumferentially, leaving necrotic skin behind the advancing edge. It is possible that individuals in current populations, in which variable levels of T cell deficiencies are seen, have less than complete absence of inflammatory response. Thus, failure to heal by 14–21 days, even with minimal inflammatory response, may characterize progressive vaccinia in individuals with lesser degrees of T cell suppression.

Virus gains entry into the blood at an early stage in patients with nearly totally deficient immune systems and implants in distant skin sites and in multiple organs. Secondary skin lesions follow the same pattern as those associated with the primary vaccination, expanding in situ. The progression is slow, and vaccinia may take weeks or months to evolve. The infection is persistent, resistant to treatment, and nearly always fatal in persons with profound T cell defects. In adult patients with lymphatic malignancies, recovery sometimes has occurred; treatment appeared to be more efficient in some of these patients [2, 3, 44, 45, 49]. It is likely that the T cell defect in these patients was of lesser degree than that in contexts such as severe combined immunodeficiency (SCID) in children. In addition, in some patients, it was possible to decrease the immune defect by temporarily limiting chemotherapy that contributed to T cell dysfunction. Local and systemic bacterial infection can ensue with progressive disease. In unsuccessfully treated patients, the condition can progress to what appears to be toxic or septic shock. Patients with T cell immunodeficiencies are susceptible to fungal and parasitic infections, and individuals with progressive vaccinia have had systemic fungal infections and infection with *Pneumocystis carinii.*

Patients with antibody deficiency but intact cell-mediated immunity (e.g., Bruton-type hypogammaglobulinemia) usually were vaccinated without incident when smallpox vaccine was routinely administered [2]. The lesions often evolved in slower fashion but resolved in most. One of the authors (V.A.F.) provided care for a patient with hypogammaglobulinemia of the Bruton type who experienced progressive vaccinia after a mumps or mumpslike infection [3]. It is likely that the viral infection depressed cell-mediated immunity and allowed a lingering vaccinia infection to become extensive. That patient survived only after massive surgical debridement, which reduced the viral mass and allowed extensive treatment with VIG and other modalities to become effective. This was the only patient

Figure 7. Progressive vaccinia in 2 children with severe combined immunodeficiency. Note the total lack of inflammatory response and the progressive, centripetal nature of spread of the lesion. (Photographs from V.A.F.)

with true hypogammaglobulinemia seen in Denver among a series of patients with progressive vaccinia during a 10-year period; all others had established T cell deficiencies (V.A.F., personal experience) [3]. Reports in the literature of hypogammaglobulinemia and progressive vaccinia are clouded by the lack of immunologic understanding at the time; many patients probably had undiagnosed CMI deficiencies.

Rarely, individuals with other disorders who were undergoing treatment with immunosuppressive agents or whose disease resulted in significant T cell suppression developed progressive vaccinia [48]. Death occurred in nearly all individuals with profound CMI defects. Some individuals survived when immune function improved coincident with the withdrawal of immunosuppressive therapy or spontaneous improvement in the underlying disease. Aggressive administration of VIG resulted in cure in such cases. Patients with milder degrees of depression of cell-mediated immunity responded to aggressive VIG therapy. In several children, graft-versus-host disease occurred as a result of blood transfusions in which viable lymphocytes that attacked the patients' tissues were infused [50].

Clinical manifestations. Most experience has been with children with varying defects in T cell immunity. The most severe disease was seen in patients with SCID with absence of both T cell and B cell function (figure 7). The first sign of progressive vaccinia is failure of the primary vaccination site to heal. Typically, the patient is not very ill at the time and does not have an inflammatory response. Lesions vary in appearance (figures 7 and 8).

Viremia ensues, and secondary viremic lesions appear anywhere on the body, each progressive in nature. Lesions may coalesce and involve large patches of skin, or an entire limb may be covered. Lesions may become necrotic, with a dark eschar covering the entire lesion. The lack of inflammation explains the absence of lymphadenopathy, hepatosplenomegaly, or any other sign or symptom of an immunologic response. In treated patients, particularly if viable lymphocytes have been given, inflammation may be seen. In less immunosuppressed individuals, some inflammatory response may be present, but healing will not occur. Bacterial superinfection is common, and if the patient has intact B cell and phagocytic function, signs of inflammation may appear, such as lymphadenopathy, splenomegaly, and erythematous and edematous reactions at the vaccinia virus–infected sites.

With progression, patients usually develop toxic or septicemic shock and disseminated intravascular coagulation. They also are subject to systemic fungal, parasitic, and opportunistic bacterial infections with septicemia and, ultimately, death. If viable unmatched lymphocytes have been administered, even with single unirradiated blood transfusions, graft-versus-host disease may occur, with splenomegaly, hepatomegaly, skin rash, disseminated intravascular coagulation, and signs of inflammatory response to vaccination sites (figure 9) [3, 50]. This syndrome has been uniformly fatal in the past.

Children with T cell deficiency but intact or nearly intact B cell function have a less devastating course, but the outcome may be the same as for children with SCID. The primary vaccination site is similar in appearance to that in children with SCID. There are few viremic lesions. Serum antibody develops, and the concomitant administration of VIG halts viremic spread. These patients also respond to surgical debridement and skin grafting aimed at decreasing the viral mass. Vigorous treatment with VIG is warranted, and, if antiviral therapy becomes available, it should be administered. Thiosemicarbazone, a weak antiviral, has been used in the past, but its value is uncertain, and it is no longer available.

Adults may have variable forms of progressive vaccinia, depending on the cause of the T cell defect, the degree of impairment, and its reversibility. The most severe disease is seen in patients who have malignancies or are undergoing chemotherapy that causes sufficient loss of CMI function to permit progressive vaccinia to occur. An example of such occurrence is seen in figure 10. This man had a lymphosarcoma requiring potent anticancer chemotherapy. He developed extensive progressive vaccinia. In another patient with lymphosarcoma, anticancer therapy was discontinued temporarily, VIG and antivirals were administered, and the patient recovered completely [3].

Diagnosis. Progressive vaccinia is diagnosed clinically by the appearance and progression of lesions at the primary vaccination site and subsequent viremic lesions. Diagnosis is aug-

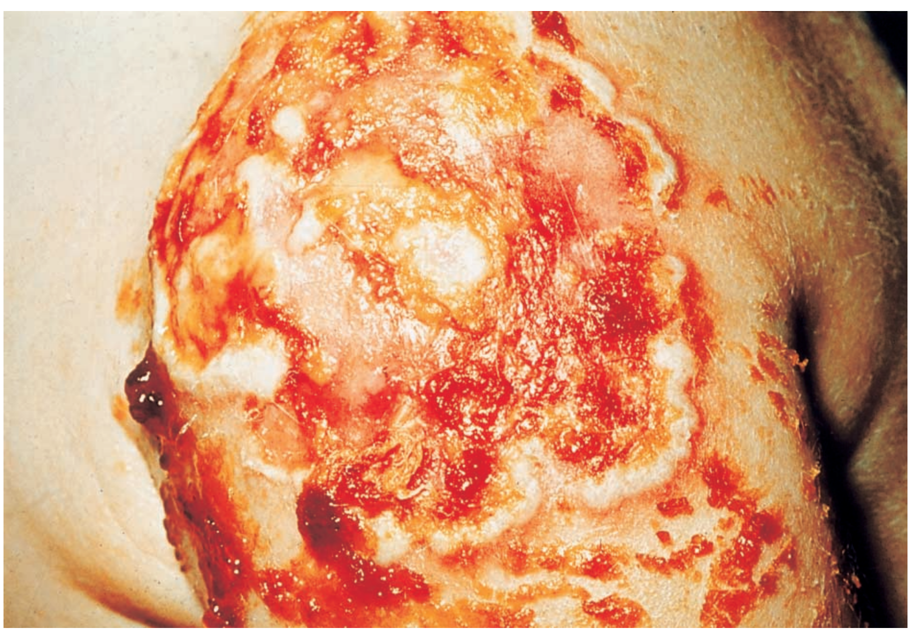

Figure 8. Progressive vaccinia. Another appearance of a progressive lesion in a child with intact antibody synthesis but lacking T cell function. (Photograph from V.A.F.)

Figure 9. Fatal graft-versus-host disease in a child with progressive vaccinia who received a blood transfusion with viable lymphocytes. Photograph was taken postmortem. (Photograph from V.A.F.)

mented by identification of the immunologic defect. In most instances, the diagnosis is clear-cut and unequivocal. The differential diagnosis includes severe bacterial infection, disseminated herpes or very severe chickenpox, and occasionally other vaccinial adverse events. Virologic and immunologic testing is essential and should be done after consultation with appropriate infectious diseases or immunology experts.

Management. True progressive vaccinia in individuals with profound T cell deficiencies is most often fatal. No successful treatment has been found; although transfer of lymphocytes has resulted in viral "cure," the resultant graft-versus-host disease was lethal [3, 50]. If blood transfusion is required in the care of patients with progressive vaccinia, the blood should be irradiated to destroy lymphocytes.

The modalities used in the past to treat this condition were administration of VIG in massive doses, administration of thiosemicarbazone, and measures such as exchange transfusions, use of irradiated blood, platelet infusions, plasma administration, specific treatment for concomitant infections by other opportunistic organisms, and surgical debridement. None of these alone or in combination were effective, except in individuals with lesser degrees of T cell defect or in adults with reversible T cell suppression.

Modern ICU-level treatment for septic/toxic shock might help some of these patients. Cidofovir (a cytosine derivative), an antiviral with some effect in vitro and in animals against poxviruses that has been licensed for use in humans with cytomegalovirus infection and other infections, has been suggested as a potential therapy for progressive vaccinia, but at present there is no evidence of its effectiveness in humans with variola or complications of vaccination [41, 42].

There is a case report of a "cure" of progressive vaccinia in an immunologically deficient infant with absent delayed-type hypersensitivity reaction, but the immunologic analyses showed some immunoglobulin production for IgG and IgM and absence of IgA, which suggests that this was one of the variants described above [51]. The child was treated minimally with VIG at a dosage of 0.5 mL/kg intramuscularly every 2 weeks for 8 weeks (4 doses) and was said to be completely healed of both local and viremic lesions. The experience of one author (V.A.F.) with 23 patients who had progressive vaccinia and were treated with VIG (plus debridement and thiosemicarbazones) only revealed "cure" in children with intact antibody capacity but deficient T cell immunity and in adults with reversible T cell dysfunction [3].

Prevention. In the absence of a smallpox outbreak, patients with T cell abnormalities should not receive smallpox vaccine. An appropriate history suggestive of T cell immunodeficiency, either on a congenital basis or secondary to some other disease or treatment, identifies a person as potentially susceptible to progressive vaccinia. Such a history would include unusual infections, such as persistent thrush; parasitic disease; and previous opportunistic bacterial infections or severe viral infections. If infant vaccinations are done, as they were in the past, a suspicious history may be lacking early in infancy; vaccinations should be deferred until 1 year of age.

Not all patients who have immune defects, cancer, or HIV infection or who are receiving immunosuppressive therapy are T cell deficient. Bartlett [52] suggested that the $CD4^+$ T cell count may be the best measure for predicting risk at this time. Experience with other infectious agents led to the selection of a $CD4^+$ T cell count of <200 cells/mm^3 as the accepted threshold for defining susceptibility.

Consultation with an immunologist is advised for patients in these categories, but prudence would dictate that they not receive vaccine in nonemergency situations. These patients also should be cautioned not to come into contact with vaccinated individuals. If a smallpox outbreak occurs, the ACIP has recommended that all patients in these categories with bona fide exposure to smallpox be vaccinated. This could result in some

Figure 10. Progressive vaccinia in an adult with lymphosarcoma. (Photograph from V.A.F.)

vaccination-related fatalities in individuals with severe T cell deficiencies. Every effort should be made to sequester these individuals to prevent exposure to smallpox and negate the need for vaccination.

Prior administration of VIG to patients subsequently exposed to smallpox has been known to reduce the frequency of smallpox, which suggests that VIG administration might prevent or ameliorate progressive vaccinia in susceptible individuals [53]. There is no evidence for this use; VIG does not correct T cell deficiency, and its use is not recommended at present.

Potential vaccinees and their close contacts should be screened for any history consistent with an immunodeficiency in the family or prior infections in the vaccinee or contact; the possibility of HIV infection or AIDS; the presence of cancer and/or chemotherapy for cancer; receipt of an organ transplant; and receipt of immunosuppressive therapy, including steroid therapy equivalent to 1–2 mg/kg of prednisone daily.

Postvaccinial Encephalitis (PVE)

General. Encephalitis or meningoencephalitis after vaccination has been reported in the United States and other countries [54–60]. A number of different infectious and noninfectious agents can produce similar disease, and it is often impossible to establish a specific etiology. For example, >100 viruses have been identified as causes of encephalitis [61].

Estimates of the frequency of PVE are given in table 2. Data indicate that the incidence of PVE was higher in Europe than in the United States, probably because more-potent vaccine strains were used in Europe or because of differences in the genetic makeup of the populations involved. In The Netherlands, the frequency of PVE in young adult military recruits was as high as 1 case/4000 vaccinees (in the United States, the rate was estimated to be 1 case/100,000 primary vaccinees [57]) There are no known predictors of susceptibility, but the incidence is somewhat higher among infants <1 year old.

The pathogenesis of this form of encephalitis is unknown. It is a para- or postinfectious encephalitis and clinically resembles what is currently termed "postvaccinal acute demyelinating encephalomyelitis." This condition probably represents some type of autoimmune process involving the white matter of the CNS. Midbrain, cerebral, and medullary lesions and transverse spinal lesions have been observed, and myelitis is predominant in one-fifth of cases. Increases in CSF pressure, CSF lymphocytosis, and increases in CSF protein content are often found. One-third of all cases reported in the past were fatal, and one-half of all survivors have had residual neurologic defects [54–60]. The high mortality rate may represent a bias of reporting in the past. Recent reports of follow-up of patients with postinfectious demyelinating encephalitis associated with a variety of agents has demonstrated a much more favorable prognosis, with no mortality and full recovery in >70% of the patients. This is in part attributable to expert ICU-level care, which was not available in the 1960s [62, 63]. In addition, 2 patients with PVE in the recent US military vaccination program had a very mild illness and recovered quickly and completely; these cases are still under investigation but were temporally associated with primary vaccinations [64].

Clinical manifestations. Encephalitis usually occurs 7–14 days after vaccination. The clinical spectrum is unclear because of the potential confusion with other causes of encephalitis, but headache, vomiting, drowsiness, and fever are frequently observed and may be all that is seen in mild cases with rapid recovery. In more-severe cases, symptoms progress to include paralysis, incontinence, urinary retention, coma, and convulsions. Death can occur suddenly, usually within 1 week of onset of symptoms; approximately one-quarter of patients die. Recovery may leave the patient without any residual effects, but a full spectrum of neurologic sequelae has been found in approximately one-third of survivors.

Diagnosis. A variety of encephalitic syndromes can mimic that attributed to vaccination, including infection with Epstein-Barr virus, herpesviruses, enteroviruses, *Mycoplasma pneumoniae,* varicella-zoster virus, and arboviruses; measles; and mumps. A variety of encephalopathies can produce similar symptoms, including multiple sclerosis in its early stages. Studies should be undertaken to rule out other causes, because the diagnosis of PVE is one of exclusion. PCR testing of CSF for a range of agents that cause meningoencephalitis should be performed. Bacterial infection should also be ruled out by examination of CSF and other appropriate cultures. WBC counts in peripheral blood and CSF are nonspecific and show an increase in mononuclear cells. CSF protein and glucose levels may be slightly elevated. MRI should provide a diagnostic image of this disorder. The MRI findings may be normal initially, but by 5–14 days after vaccination, characteristic multifocal lesions are present throughout the white matter [65].

Typically, temporal association with vaccination (onset within 7–14 days, on average) and meningoencephalitic symptoms suggest this entity. Some authors have recovered vaccinia virus from CSF [55, 60], but researchers in many of the studies cited have been unable to do so [56, 58]. The usual methods (electroencephalogram, CT, and MRI and virologic and antibody studies of CSF) should be used to differentiate acute infectious encephalitis from postinfectious encephalitis.

Management. There is no specific therapy for PVE. Supportive care, anticonvulsants, and intensive care may be required in individual cases. In instances of acute demyelinating encephalopathy that is not caused by vaccination, the use of steroids has been indicated. It is probably unwise to consider use of steroids to treat PVE because of the potential adverse effect on the immune response to vaccinia. VIG is not effective in treating PVE, and its use is not recommended.

Prevention. There are no specific indicators of susceptibility to PVE, although the incidence is modestly increased among infants. Patients with an evolving CNS disorder should not be vaccinated. A history of convulsions is not known to be associated with a higher frequency of PVE. Stable CNS disorders should not be a contraindication to vaccination. In The Netherlands, the use of 2 mL of VIG, administered at the time of vaccination, appeared to significantly reduce the subsequent incidence of PVE among military recruits [57]. The vaccine used in The Netherlands was the Lister strain, which is more reactogenic than that used in the United States. Prophylactic use of VIG is not recommended in the United States.

Other Alleged Adverse Reactions

From time to time in the past, individual illnesses have been noted in temporal association with smallpox vaccination. These include, but are not limited to, myocarditis, pericarditis, hemolytic anemia, various neurologic disorders, melanoma in the vaccination scar, and osteomyelitis. These reports did not present sufficient evidence to establish a causal relationship between the illness and the vaccination, and therefore many experts have discounted an etiologic association. Among these, myocarditis and pericarditis have been noted in the interval after smallpox vaccination [2]. Whether there is an etiologic association for these reported temporal associations is uncertain.

At this time, it is not possible to include any adverse reactions attributable to smallpox vaccination other than those listed in this review. The CDC plans careful surveillance and epidemiologic investigation as more vaccinations are performed in the United States, and we may learn whether the spectrum of adverse events still includes only those included in this report or needs to be expanded.

VIG

General. Studies in the 1950s and 1960s indicated that complications of vaccination appeared to occur soon after vaccination and before significant levels of antibody could be detected in the blood. As a result, a method of providing antibody in the form of γ-globulin was developed by Kempe and others [66, 67]. Serum samples from hyperimmunized military personnel were harvested, and concentrated γ-globulin was prepared as a 16.5% solution. VIG was used to treat conditions that previously had no effective treatment, and such treatment resulted in improvement in prognoses and reduction in mortality. This led to the formulation of a standard preparation of VIG for intramuscular use [68]. VIG was initially distributed by the American Red Cross, after consultation with one of a series of recruited experts. The distribution and consultation system was transferred to the CDC for more uniformity and systematic record-keeping [69].

Empiric but uncontrolled evidence suggests that patients with EV, some patients with severe generalized vaccinia, many patients with severe reactions to inadvertent inoculation, and some patients with milder forms of progressive vaccinia can benefit from VIG. Smallpox vaccination ceased in the early 1970s, before definitive studies had been done to determine the efficacy of VIG. In the British experience, 90% of adverse events that occurred after vaccination appeared to respond to VIG (including EV, generalized vaccinia, and reactions to accidental inoculation).

Intramuscular VIG was produced in the 1960s from plasma obtained from recently vaccinated donors. It contained a high titer of anti–vaccinia virus neutralizing antibody. It was administered solely by the intramuscular route and could not be used intravenously for fear of causing anaphylactic reactions, because it contained a high proportion of aggregated protein.

With the reinstitution of vaccination against smallpox, there is increased interest in the use of VIG. Intramuscular VIG, which was released as a licensed product in 1995, has been stored at the CDC and is available under investigational new drug protocols. An effort is under way to produce new lots that will meet the standards for intravenous VIG (IV-VIG). By the summer of 2004, sufficient amounts of IV-VIG will be available to treat anticipated complications, should it be necessary to vaccinate 300 million persons. It is anticipated that only IV-VIG will be used for any adverse events requiring therapy.

VIG is not recommended for treatment of vaccinia keratitis, mild instances of accidental inoculation, erythematous rashes that occur after vaccination ("erythema multiforme"), Stevens-Johnson syndrome, or PVE.

Dose of VIG. In the period from the 1950s through the 1970s, the recommended dose of VIG varied. An initial dose of 0.6 mL/kg of body weight was injected intramuscularly, and subsequent administration was determined by the course of the illness. In severe cases of EV and progressive vaccinia, doses of as much as 1–10 mg/kg were used. These large doses were split into smaller units and injected at multiple sites over time. In one patient with progressive vaccinia, the intramuscular preparation of VIG was diluted in blood used in exchange transfusion and administered intravenously without serious consequences, but also without discernible beneficial effect. The recommended dosage of IV-VIG is currently being established by clinical trials. Consult http://www.bt.cdc.gov/training/smallpoxvaccine/reactions/vig_reference.html for up-to-date information on dosage and availability.

Current status. There is sufficient intramuscular VIG at the CDC to treat 600–800 adverse events. New lots of IV-VIG have been and are continuing to be produced that conform to standard for intravenous use. This IV-VIG will require new recommendations for dosage and method of administration. The new IV-VIG has a low level of aggregated protein, which allows it to be used by either the intramuscular or the intravenous route. All VIG is considered to be an investigational drug and must be used under investigational new drug protocols. IV-VIG most likely will be administered at a lower dose than the intramuscular preparation. In preliminary laboratory studies of animal models, IV-VIG appears to be 5–10 times as potent as the intramuscular preparation.

SUMMARY

Although this report has focused on the major known adverse events that occur after smallpox vaccination, we want to emphasize the safety and effectiveness of smallpox vaccination for almost all recipients. Factors that predict the possibility of susceptibility to specific adverse events can be identified, and attention to these factors in interviewing potential vaccinees before vaccination can prevent vaccination and avoid adverse events. In the absence of a smallpox outbreak, many individuals should not receive smallpox vaccine because of the presence of potential susceptibilities. If smallpox is introduced into the general population, however, many of these contraindications will no longer apply.

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